YOLOV10

Overview:

1. Used BDD100K Dataset that includes 10000 images and 12 classes
2. Ten classes used: bike, bus, car, motor, person, rider, traffic light, traffic sign, train, truck.
3. Extracted 1200 samples from the dataset out of which 1000 and 200 are train and validation set respectively.

   Code:

```python
import os
import shutil

#Source path
src_img_dir = "D:\\Smart vision\\BDD100K\\bdd100k\\bdd100k\\images\\100k\\val" #Images
src_lb_dir = "D:\\Smart vision\\BDD100K\\bdd100k_labels_release\\bdd100k\\labels\\validation" #labels

#Target path
tar_img_dir = "D:\\SmartVision (Computer Vision)\\Sample train-12000\\images\\val" #Images
tar_lb_dir =  "D:\\SmartVision (Computer Vision)\\Sample train-12000\\labels\\val" #labels

images_file = [f for f in os.listdir(src_img_dir) if f.endswith(('jpg','png'))]

#Extract 5000 images
extracted_image = images_file[:200]

for img in extracted_image:
    base = os.path.splitext(img)[0]

    #Source
    img_path = os.path.join(src_img_dir, img)
    label_path = os.path.join(src_lb_dir, base+'.txt')

    #Destination
    tar_img_path = os.path.join(tar_img_dir, img)
    tar_lb_path = os.path.join(tar_lb_dir, base+'.txt')

    #Copying Image
    shutil.copy(img_path, tar_img_path)

    if os.path.exists(label_path):
        shutil.copy(label_path, tar_lb_path)
    else:
        print(f"Warning: No label found for {img}")
```

4. Annotated images saved in *.json* format are converted to *.txt* yolo compatible format:

[Class_id] [x_center] [y_center] [width] [height]

For example,

```
9 0.491778 0.432859 0.009186 0.009526
2 0.274377 0.504889 0.039806 0.057157
2 0.227299 0.502167 0.071191 0.078931
2 0.168355 0.518498 0.081908 0.087097
```

Code:

```python
import json
import os

#File path to BDD100K Train file
file_path_train = 'D:\\Smart vision\\BDD100K\\bdd100k_labels_release\\bdd100k\\labels\\bdd100k_labels_images_train.json'

#File path to label file
output_label_path = 'D:\\Smart vision\\BDD100K\\bdd100k_labels_release\\bdd100k\\labels\\train'

#Labelled_classs
classes = ['bike', 'bus', 'car', 'drivable area', 'lane', 'motor', 'person', 'rider', 'traffic light', 'traffic sign', 'train', 'truck']

os.makedirs(output_label_path, exist_ok = True)

#Usual dimensions of the BDD100K dataset images
image_width = 1280
image_height = 720

#Function that computes the Normalized co-ordinates of the center, width and height
def convert_box(box):
    x1, y1, x2, y2 = box['x1'], box['y1'], box['x2'], box['y2']
    x_center = (x1 + x2) / 2 / image_width
    y_center = (y1 + y2) / 2 / image_height
    width = (x2 - x1) / image_width
    height = (y2 - y1) / image_height
    return x_center, y_center, width, height

#Loading .json file
with open(file_path_train,"r") as f:
    data = json.load(f)

for i, item in enumerate(data):
    image_name  = item['name']
    labels = item['labels']

    #Create Output file .txt with the same base name as the real train image
    label_file_path = os.path.join(output_label_path, image_name.replace('.jpg','.txt'))

    with open(label_file_path,"w") as label_file:
        for obj in labels:
            category = obj.get('category')
            if category not in classes:
               continue # Skips if the class is unknown
            class_id = classes.index(category)
            if 'box2d' not in obj:
               continue # Skips if there is no bounding boxes
            x , y, w, h = convert_box(obj['box2d'])
            label_file.write(f"{class_id} {x:.6f} {y:.6f} {w:.6f} {h:.6f}\n")

    if i % 1000 == 0:
        print(f"{i} images processed successfully...") # ~69000 Images processed

print("Conversion completed. YOLO labels saved to: ", output_label_path)
```

5. Then, created .yaml file with train and val image path along with number and name of the classes.
6. Then, the model is trained with the following parameters:

```python
[ ] from ultralytics import YOLO

    # Load YOLOv10n model from scratch
    model = YOLO("yolov10n.pt")

    # Train the model
    model.train(data="/content/Sample train-1200/bdd.yaml", epochs=30, batch=8, imgsz=640, visualize=True)
```

Parameter:

- Model: YOLOv10 (Nano)
- Dataset: BDD100k (1200 samples)
- Epochs: 30
- Batch: 8
- Input size: 640 x 640

7. Sample output(**Demo in the classroom**)

8. <u>Key challanges:</u>
   - Annotation Format Conversion
   - Label Imbalance
   - Parameters like batch, imgsz, and epochs significantly affect performance.